

Ideation Phase — Brainstorm & Idea Prioritization Report

Software Installation Request Automation — ServiceNow Service Catalog Project

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Project Title	Software Installation Request Automation via ServiceNow Service Catalog

1. Project Overview

The Software Installation Request Automation project in ServiceNow streamlines how employees request, get approval for, and receive software installations across the organization. Traditionally, such requests travel through emails or verbal communication with IT staff, which creates inconsistent approval trails, delayed fulfillment, poor visibility into request status, and software license compliance risks. This project replaces that ad-hoc process with a standardized, auditable Service Catalog item backed by automated workflows spanning approval, license validation, and deployment.

By digitizing the request lifecycle end-to-end, the organization gains a single system of record for every software request, enforces consistent approval policies, and reduces the manual effort IT teams spend tracking installations and licenses.

2. Step 1 — Team Gathering & Problem Statement

2.1 Problem Statement

Software installation requests are commonly processed through emails or verbal communication, causing inconsistent approvals, delayed fulfillment, lack of request visibility, and software license compliance issues. This project proposes a standardized ServiceNow Service Catalog solution that automates request submission, approvals, license verification, task creation, and fulfillment.

2.2 How Might We?

How might we design a standardized ServiceNow Service Catalog workflow that automates software approvals, license validation, and fulfillment while eliminating manual delays and compliance risks?

2.3 Session Goals

- Replace manual, email-based requests with a structured Service Catalog item.
- Automate approvals using Flow Designer, including manager and risk-based approval paths.
- Integrate the catalog workflow with Software Asset Management (SAM) for license tracking.
- Enable automated deployment through IntegrationHub, SCCM, or Microsoft Intune.
- Improve transparency using request tracking, notifications, and self-service status visibility.

3. Step 2 — Idea Listing & Grouping

Ideas generated during the ideation session were grouped into three functional areas covering the catalog experience, the underlying workflow logic, and the fulfillment mechanics.

3.1 Group A — Service Catalog

- **A1 — Dynamic Catalog Item:** Custom-built catalog item that dynamically shows software options based on department, role, or OS platform.
- **A2 — License Availability Indicator:** Real-time indicator on the catalog item showing whether licenses are currently available before submission.
- **A3 — Self-Service Status Tracker:** A self-service portal widget where employees can track the live status of their software request without contacting IT.

3.2 Group B — Workflow

- **B1 — Tiered Manager Approval:** Multi-level manager approval chain that scales based on software cost or licensing tier.
- **B2 — Timeout Escalation:** Automatic escalation to the next approver or IT admin if an approval is not actioned within a defined SLA window.
- **B3 — Dynamic Risk Assessment:** Flow logic that evaluates the requested software against a risk/security catalog and routes high-risk software through an additional security review.

3.3 Group C — Fulfillment

- **C1 — SAM Integration:** Direct integration with the Software Asset Management application to decrement/allocate license counts automatically on approval.
- **C2 — Intune/SCCM Deployment:** IntegrationHub spokes connecting to Microsoft Intune or SCCM to push a zero-touch software installation to the employee's device.
- **C3 — Automated Ticket Closure:** Automatic closure of the Requested Item (RITM) and parent Request (REQ) once the fulfillment task confirms successful installation.

4. Step 3 — Idea Prioritization

Each idea was plotted against Importance (business/user value) and Feasibility (ease of implementation within the current ServiceNow instance and available integrations) to decide what belongs in the Minimum Viable Product (MVP) versus later phases.

Priority Quadrant	Ideas
High Importance / High Feasibility	Dynamic Catalog Item, Tiered Manager Approval, Automated RITM Closure
High Importance / Low Feasibility	SAM Integration, Zero-Touch (Intune/SCCM) Deployment
Low Importance / High Feasibility	Self-Service Status Tracker, Timeout Escalation
Low Importance / Low Feasibility	Real-Time License Availability Indicator

MVP Scope: The High Importance / High Feasibility items form the core MVP — a dynamic catalog item, tiered manager approval, and automated ticket closure — since they deliver the greatest immediate reduction in manual effort with the least implementation risk. High-value but low-feasibility items such as SAM integration and zero-touch deployment are targeted for Phase 2 once the core flow is stable, as they depend on external system connectivity (SAM data model, Intune/SCCM APIs).

5. Proposed Solution Architecture

The end-to-end flow moves a request from submission to fulfillment without manual intervention, branching only where a licensing or approval decision is required:

Service Portal Catalog Item → Flow Designer Trigger → Software Type Check → Free Software (Skip Approval) / Licensed Software (Manager Approval) → SAM License Allocation → IntegrationHub (Intune/SCCM) → Software Installed → Request Item Automatically Closed.

5.1 Architecture Stage Breakdown

- **Service Portal Catalog Item:** The employee selects the desired software from a Service Catalog item exposed on the Service Portal.
- **Flow Designer Trigger:** Submission of the Requested Item (RITM) fires a Flow Designer flow that governs the rest of the lifecycle.
- **Software Type Check:** The flow inspects catalog variables to classify the software as free/open-source or licensed/paid.
- **Approval Branch:** Free software proceeds directly to fulfillment; licensed software is routed to the requester's manager (and, for high-cost items, a secondary approver) for sign-off.
- **SAM License Allocation:** On approval, the Software Asset Management application checks and allocates an available license before proceeding.
- **IntegrationHub Deployment:** An IntegrationHub spoke calls Microsoft Intune or SCCM to silently push and install the software on the employee's endpoint.
- **Automated Closure:** A confirmation callback marks the catalog task complete, closes the RITM and REQ, and notifies the employee via email/portal notification.

6. Expected Benefits

- Faster turnaround from request to installed software through automated routing and deployment.
- Full audit trail of every approval, license allocation, and installation for compliance reporting.
- Reduced IT workload by eliminating manual license lookups and ticket updates.
- Better license utilization visibility through direct SAM integration.
- Improved employee experience via self-service submission and status tracking.

7. Next Steps

- Build the Service Catalog item and variable set for software selection in a Personal Developer Instance (PDI).
- Design the Flow Designer workflow covering approval, SAM allocation, and fulfillment task creation.
- Configure IntegrationHub connectivity for Intune/SCCM (or simulate with a mock fulfillment task for demo purposes).
- Test end-to-end with both free and licensed software scenarios, including approval timeout escalation.
- Document the solution and prepare a walkthrough demo for evaluation.